

## FELKER AAF, VA, USA

WMO: 723087

Lat: 37.133N

Lon: 76.600W

Elev: 12

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 93735

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 19.9       | 23.7 | 1.3                           | 5.9 | 22.9 | 6.3 | 7.6 | 27.1 | 22.7                  | 33.1 | 19.7 | 34.7 | 7.3                   | 10   | 0.402 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.0                   | 93.6            | 78.2 | 91.3 | 77.5 | 89.3 | 76.4 | 82.6                | 87.1 | 80.7 | 86.4 | 79.2 | 85.5 | 6.6                  | 220  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 81.8                            | 165.6 | 83.7 | 79.3 | 152.3 | 82.8 | 77.4 | 142.6 | 82.1 | 46.4          | 87.3 | 44.2 | 86.3 | 42.6 | 85.6 | 89.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 18.4              | 15.9 | 12.9 | 15.1                       | 97.5 | 4.4                | 2.8 | 11.9                                               | 99.5 | 9.3        | 101.2 | 6.8        | 102.7 | 3.6        | 104.8 |
|                   |      |      | 12.5                       | 84.0 | 3.8                | 3.4 | 9.8                                                | 86.5 | 7.6        | 88.5  | 5.4        | 90.4  | 2.6        | 92.9  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |       |            |       |            |       |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 61.1  | 39.6  | 43.6  | 49.5  | 60.3  | 69.3  | 77.0  | 80.5  | 79.0  | 74.0  | 63.4  | 51.0  | 45.3 | (6)  |
| (7)  |                                                                   | DBStd       | 16.15 | 9.52  | 10.00 | 9.55  | 8.21  | 6.94  | 5.07  | 4.20  | 3.79  | 5.53  | 7.77  | 7.70  | 8.94 | (7)  |
| (8)  |                                                                   | HDD50       | 1000  | 344   | 229   | 133   | 10    | 0     | 0     | 0     | 0     | 3     | 79    | 202   | (8)  |      |
| (9)  |                                                                   | HDD65       | 3267  | 788   | 601   | 486   | 188   | 41    | 1     | 0     | 3     | 123   | 424   | 612   | (9)  |      |
| (10) |                                                                   | CDD50       | 5064  | 21    | 49    | 117   | 320   | 598   | 811   | 945   | 900   | 719   | 419   | 108   | 57   | (10) |
| (11) |                                                                   | CDD65       | 1856  | 0     | 1     | 5     | 48    | 174   | 362   | 480   | 435   | 272   | 74    | 3     | 2    | (11) |
| (12) |                                                                   | CDH74       | 15990 | 0     | 8     | 37    | 358   | 1212  | 3215  | 4893  | 3921  | 1909  | 426   | 9     | 2    | (12) |
| (13) |                                                                   | CDH80       | 5754  | 0     | 0     | 2     | 81    | 338   | 1199  | 2038  | 1438  | 574   | 83    | 1     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 6.0   | 6.4   | 6.8   | 7.2   | 7.0   | 6.0   | 5.5   | 5.4   | 5.0   | 5.5   | 5.8   | 5.8   | 5.7  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 46.9  | 3.3   | 2.9   | 3.8   | 3.5   | 3.7   | 4.1   | 5.5   | 5.3   | 4.8   | 3.8   | 2.8   | 3.4  | (15) |
| (16) |                                                                   | PrecMax     | 66.7  | 6.3   | 7.5   | 8.7   | 7.0   | 6.1   | 8.3   | 9.1   | 11.2  | 19.1  | 8.9   | 8.9   | 7.6  | (16) |
| (17) |                                                                   | PrecMin     | 32.7  | 1.1   | 1.1   | 0.2   | 1.4   | 0.7   | 1.4   | 1.7   | 0.3   | 0.0   | 0.1   | 1.2   | (17) |      |
| (18) |                                                                   | PrecStd     | 8.1   | 1.4   | 1.5   | 1.7   | 1.3   | 1.3   | 1.6   | 1.9   | 2.7   | 4.0   | 2.0   | 1.9   | 1.5  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 68.7  | 75.2  | 78.1  | 87.1  | 90.3  | 96.0  | 97.5  | 94.4  | 92.2  | 86.3  | 75.0  | 72.7 | (19) |
| (20) |                                                                   | 0.4%        | MCWB  | 62.2  | 63.9  | 62.8  | 70.2  | 75.3  | 77.9  | 80.5  | 79.4  | 74.7  | 74.1  | 65.1  | 68.1 | (20) |
| (21) |                                                                   | 2%          | DB    | 63.5  | 69.6  | 73.6  | 82.1  | 86.5  | 92.1  | 94.4  | 91.9  | 89.1  | 82.3  | 70.6  | 68.0 | (21) |
| (22) |                                                                   | 2%          | MCWB  | 56.9  | 59.1  | 60.3  | 66.6  | 73.2  | 77.2  | 79.1  | 78.1  | 75.3  | 71.8  | 63.9  | 64.1 | (22) |
| (23) |                                                                   | 5%          | DB    | 59.5  | 65.0  | 69.4  | 78.2  | 84.0  | 89.3  | 91.8  | 89.6  | 86.4  | 79.0  | 67.0  | 63.9 | (23) |
| (24) |                                                                   | 5%          | MCWB  | 54.1  | 57.0  | 58.2  | 65.6  | 72.2  | 76.4  | 78.3  | 76.3  | 74.5  | 69.5  | 60.7  | 60.5 | (24) |
| (25) |                                                                   | 10%         | DB    | 55.0  | 60.4  | 65.3  | 74.5  | 80.9  | 86.9  | 89.4  | 87.3  | 83.4  | 75.5  | 63.8  | 59.7 | (25) |
| (26) |                                                                   | 10%         | MCWB  | 50.0  | 53.4  | 57.6  | 64.4  | 70.3  | 75.0  | 77.5  | 75.5  | 73.3  | 68.6  | 59.2  | 56.1 | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 63.6  | 65.7  | 66.2  | 74.5  | 77.5  | 82.9  | 86.3  | 83.6  | 81.4  | 77.2  | 69.3  | 70.4 | (27) |
| (28) |                                                                   | 0.4%        | MCDWB | 66.6  | 72.0  | 73.1  | 77.6  | 83.4  | 86.5  | 88.8  | 88.6  | 82.9  | 81.7  | 70.6  | 70.8 | (28) |
| (29) |                                                                   | 2%          | WB    | 59.2  | 62.5  | 63.7  | 70.4  | 75.4  | 80.2  | 83.9  | 81.3  | 78.7  | 74.7  | 65.7  | 64.0 | (29) |
| (30) |                                                                   | 2%          | MCDWB | 63.0  | 67.7  | 69.3  | 77.8  | 83.3  | 86.3  | 87.4  | 86.7  | 83.3  | 79.3  | 68.6  | 66.7 | (30) |
| (31) |                                                                   | 5%          | WB    | 54.6  | 59.0  | 61.5  | 67.9  | 73.9  | 78.4  | 81.5  | 79.8  | 77.3  | 72.6  | 62.9  | 61.0 | (31) |
| (32) |                                                                   | 5%          | MCDWB | 58.4  | 64.1  | 67.6  | 76.0  | 81.1  | 85.6  | 87.5  | 85.7  | 82.3  | 77.1  | 65.9  | 63.8 | (32) |
| (33) |                                                                   | 10%         | WB    | 50.4  | 53.8  | 58.4  | 65.2  | 72.2  | 77.2  | 79.6  | 78.3  | 75.9  | 69.7  | 59.6  | 56.7 | (33) |
| (34) |                                                                   | 10%         | MCDWB | 54.2  | 59.0  | 64.2  | 73.1  | 78.7  | 84.5  | 86.5  | 84.3  | 80.7  | 74.4  | 63.4  | 59.0 | (34) |
| (35) | Mean Daily Temperature Range                                      |             | MDBR  | 15.7  | 17.6  | 17.7  | 18.3  | 16.7  | 16.2  | 16.0  | 15.2  | 15.0  | 16.5  | 16.6  | 15.5 | (35) |
| (36) |                                                                   | 5% DB       | MCDBR | 22.8  | 24.0  | 23.6  | 22.0  | 19.5  | 18.7  | 18.6  | 17.7  | 17.9  | 19.1  | 20.4  | 18.2 | (36) |
| (37) |                                                                   |             | MCWBR | 18.7  | 17.0  | 14.1  | 10.8  | 9.6   | 8.4   | 7.6   | 7.6   | 8.3   | 10.1  | 14.0  | 16.4 | (37) |
| (38) |                                                                   |             | MCDBR | 20.3  | 23.5  | 19.4  | 19.0  | 16.9  | 17.0  | 17.0  | 15.9  | 14.9  | 15.7  | 17.4  | 17.8 | (38) |
| (39) |                                                                   | 5% WB       | MCWBR | 18.7  | 18.7  | 14.5  | 11.5  | 9.0   | 9.8   | 9.3   | 8.0   | 8.5   | 9.7   | 14.9  | 16.8 | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.305 | 0.318 | 0.346 | 0.392 | 0.434 | 0.487 | 0.516 | 0.511 | 0.409 | 0.354 | 0.322 | 0.309 | (40) |      |
| (41) |                                                                   | taud        | 2.537 | 2.497 | 2.450 | 2.331 | 2.244 | 2.147 | 2.082 | 2.083 | 2.357 | 2.507 | 2.547 | 2.559 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 283   | 292   | 292   | 282   | 271   | 255   | 247   | 245   | 267   | 275   | 273   | 273   | (42) |      |
| (43) |                                                                   | Edh at Noon | 24    | 29    | 33    | 40    | 44    | 48    | 51    | 50    | 36    | 28    | 24    | 23    | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 744   | 981   | 1304  | 1655  | 1868  | 2013  | 1918  | 1730  | 1404  | 1113  | 839   | 660   | (44) |      |
| (45) |                                                                   | RadStd      | 60    | 108   | 105   | 122   | 167   | 134   | 94    | 91    | 122   | 131   | 69    | 53    | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |      |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |      |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |      |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   | (46) |
| (47) | Regional (70 neighbors) | +0.85 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +299  | +191  | (47) |

Nomenclature: See separate page